Name: Qussay Suleiman

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CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, SPRING 2025, CLOSED BOOK, TIME: 60 MIN

The endothermic gas phase reaction $A \rightarrow B + C$ is carried out in a single steam-jacketed fluidized bed reactor, where the temperature of the steam circulating through the jacket remains constant at $T_a = 300°C$. 573.15 K

For the following data, calculate the steady-state reactor temperature.

Conversion: $X = 85\%$

Heat-transfer area of the jacket: $A = 5\ m^2$

Overall heat transfer coefficient of the jacket: $U = 500 \frac{J}{s.m^2.K}$

Heat of reaction per unit mole of species A at 25°C: $\Delta_r H = 120 \frac{kJ}{mol}$ = $120000 \frac{J}{mol}$

Feed temperature: $T_0 = 40$ °C 313.15 K 298.15 K

Feed molar flow rates: $F_{A0} = 10 \frac{mol}{s}$, $F_{I0} = 2 \frac{mol}{s}$ (inert), $F_{B0} = F_{C0} = 0$ $\theta_A = 1,\ \theta_I = \frac{2}{10} = \frac{1}{5},\ \theta_B = \theta_C = 0$

Heat capacities (assume constant) $c_{pA} = 90 \frac{J}{mol.K}$, $c_{pB} = 60 \frac{J}{mol.K}$, $c_{pC} = 40 \frac{J}{mol.K}$, $c_{pI} = 50 \frac{J}{mol.K}$

Reactor energy balance: $\frac{dU}{dt} = \sum_\alpha F_{\alpha 0}\, h_\alpha(T_0) - \sum_\alpha F_\alpha\, h_\alpha(T) + \dot{Q} + \dot{W}_s$

$\Delta_r C_p = \underbrace{40}_{C} + \underbrace{60}_{B} - \underbrace{90}_{A} = -10 \frac{J}{mol.K}$

Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$

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$$T = \frac{\left[(298.15\,K)(-10 \frac{J}{mol.K}) - 120000 \frac{J}{mol}\right](0.85) + (313.15\,K)(115 \frac{J}{mol.K}) + \frac{(500 \frac{J}{s.m^2.K})(5m^2)(573.15\,K)}{10 \frac{mol}{s}}}{115 \frac{J}{mol.K} + (0.85)(10 \frac{J}{mol.K}) + \frac{(500 \frac{J}{s.m^2.K})(5m^2)}{10 \frac{mol}{s}}}$$

units:

$[\ ] = \frac{J/mol}{J/mol.K} = K$

$T = 213.746\ K$ or $-59.4°C$

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E.B: $\frac{d\cancel{U}^{0}}{dt} = \sum_a F_a h_a(T_0) - \sum_a F_a h_a(T) + \overset{-Ua(T-T_a)}{\dot{Q}}$

$$0 = \sum_a F_a h_a(T_0) - \sum_a (F_{a0} + \nu_a X) h_a(T) - Ua(T-T_a)$$

$$0 = \sum_a F_a[h_a(T_0) - h_a(T)] - F_{A0} X \sum \nu_a h_a(T) - Ua(T-T_a) = 0$$

$$\sum_a \theta_a C_{pa}(T_0 - T) - X \overbrace{\Delta_r H(T_r)}^{\Delta_r H(T_R) + \Delta_r C_p(T-T_R)} - \frac{Ua}{F_{A0}}(T - T_a) = 0$$

$$\cancel{T = \frac{[T_R \Delta_r C_p - \Delta_r H(T_R)]X + T_0 \sum_a \theta_a C_{pa}}{\sum_a \theta_a C_{pa} + \Delta_r C_p X}}$$

$$\sum_a \theta_a C_{pa}(T_0 - T) - \frac{Ua}{F_{A0}}(T - T_a) = X(\Delta_r H(T_R) + \Delta_r C_p (T - T_R))$$

$$-T\sum_a \theta_a C_{pa} + T_0 \sum_a \theta_a C_{pa} - \frac{Ua}{F_{A0}}(T - T_a) - X\Delta_r C_p T = X(\Delta_r H(T_R) - T_R \Delta_r C_p)$$

$$-T\sum_a \theta_a C_{pa} - \frac{Ua T}{F_{A0}} + \frac{Ua T_a}{F_{A0}} - X\Delta_r C_p T = X(\Delta_r H(T_R) - T_R \Delta_r C_p) - T_0 \sum_a \theta_a C_{pa}$$

$$-T\left(\sum_a \theta_a C_{pa} + \frac{Ua}{F_{A0}} + X\Delta_r C_p\right) = X(\Delta_r H(T_R) - T_R \Delta_r C_p) - T_0 \sum_a \theta_a C_{pa} - \frac{Ua T_a}{F_{A0}}$$

$$T = \frac{[T_R \Delta_r C_p - \Delta_r H(T_R)]X + T_0 \sum_a \theta_a C_{pa} + \frac{Ua}{F_{A0}} T_a}{\sum_a \theta_a C_{pa} + X \Delta_r C_p + \frac{Ua}{F_{A0}}}$$

$$\sum_a \theta_a C_{pa} = \overbrace{(1)(90)}^{A} + \overbrace{(\tfrac{1}{5})(50)}^{I} + \overbrace{(0)(60)}^{B} + \overbrace{(0)(40)}^{C} = 115 \frac{J}{mol \cdot K}$$

computation on cover sheet cause of lack of space